Information Gain Based Informative Path Planning for UAVs in Agriculture: Towards Field Deployment

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Abstract. Precision agriculture relies on accurate and efficient data collection to optimise crop management. Wheat lodging, a critical issue that affects yield and quality, can be assessed using Unmanned Aerial Vehicles (UAVs) equipped with down-facing cameras. To ensure sufficient efficiency and accuracy, we propose an informative path planning (IPP) approach based on the computation of the expected information gain (IG) from future observations, allowing non-uniform monitoring of the field with the UAV flying at varying altitudes. To move towards field deployment, abstract sensor models must be replaced by models representing the actual feature detection process. Here, we consider a classifier for wheat lodging implemented through a trained deep neural network, and propose an empirically derived sensor model obtained from minimal sampling of the target field. We validate our method using both simulated and real-world wheat fields, demonstrating its effectiveness in mapping. This work bridges the gap between real-world image data and simulated IPP, opening to actual deployment of practical solutions for precision agriculture.

Keywords: UAV · Informative path planning · Precision agriculture · Wheat lodging

1 Introduction

Unmanned Aerial Vehicles (UAVs) can have a large impact in precision agriculture as they can effectively collect field data to support various tasks such as pest management, fertilisation/irrigation management, crop health monitoring, yield forecasting, harvesting and post-harvest stages [12,19]. These tasks often require accurate imagery at very high resolution to support feature detection, usually

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F. Mastrogiovanni and F. Amigoni (Eds.): IAS 2025, LNNS 2103, pp. 298–313, 2027.
https://doi.org/10.1007/978-3-032-33874-7_22

executed through some machine learning (ML) approach [17]. There is therefore a tension between the need to reduce the flight altitude to obtain images at a sufficient resolution and the need to quickly cover large fields, which calls for high-altitude passages to widen the camera footprint. In fact, the main limitations acknowledged to UAVs in precision agriculture are the lack of autonomy and the number of operations required to cover a large field [19].

A shining example of precision agriculture tasks that require precise information over vast fields is assessing lodging in staple cereal crops (e.g., wheat, barley). Lodging is the bending or breaking of stems, and can significantly impact crop yield and quality [3]. It occurs due to environmental factors, poor agronomic practices, soil conditions, and pest or disease pressure.

Detection and mapping of lodged areas is crucial for effective crop management, as well as for settlement of agricultural insurance claims. As a matter of fact, remote sensing approaches are largely studied, often relying on UAVs for imagery collection due to the high resolution required for precise assessments [1,3,8,20]. Considering the usually very large size of fields, however, assessing lodging by means of UAV imagery remains a costly exercise if approached with uniform coverage approaches, such as traditional lawnmower paths.

To obviate these limitations, non-uniform coverage strategies based on informative path planning (IPP) have been proposed, as a method to increase flight autonomy and mapping efficiency [7,14]. IPP dynamically refines observation sequences based on field data, adapting the flight path based on actual observations of the field, e.g., by means of receding horizon planners [21], Monte Carlo tree search [9], and optimisation-based approaches [5,14,18]. These strategies allow us to iteratively reduce uncertainty by enabling multiple observations, leveraging information-theoretic measures like the information gain (IG) [10,13]. Multi-UAV strategies improve mapping efficiency by partitioning fields or by coordinating action in high-interest regions [2,13,15].

While being very promising, IPP approaches have not yet been fully demonstrated for field operation. Several studies are limited to abstract simulations [7,13], others arrive at validation with a proof-of-concept [14]. Crucially, the main limiting factor is represented by the need of estimating onboard the quality of the performed observations or the residual uncertainty in order to plan effective and informative paths. Abstract sensor models, such as exponential decay functions based on observation altitude, do not grasp the actual complexity of real-world feature detection processes [13,14]. Approaches that learn the relationship between field data and sensor accuracy require annotated datasets, which may not be available for field operations [14,18]. Field deployment requires tailored and inexpensive methodologies to account for the conditions of the specific field monitoring task.

In this paper, we move a crucial step towards field deployment of IPP with UAVs for agricultural field mapping. Our approach is tailored to classification problems, and specifically we consider the need to determine if a given portion of a wheat field is lodged or not. However, extension to different problems is straightforward. We assume that a classification model is available—e.g., a

suitably trained convolutional neural network (CNN). The classification model represents the sensor used for observation and mapping, and its classification errors depend on several factors related to the specific field, the actual weather conditions and the observation altitude. Hence, we propose that an empirically derived sensor model is constructed for the specific field/day of observation with minimal targeted sampling of the field. The first step is the collection of a few images at different altitudes (i.e., where lodged areas are visible). Then, manual annotation of clearly identifiable lodged areas is performed. These annotated images are exploited to estimate the error rates of the classification model at different altitudes, therefore providing a confusion matrix that replaces the abstract sensor model in the IPP approach. We show that thanks to such an empirically supported IPP approach, the mapping accuracy remains high even with a very limited sampling effort.

Our contributions are as follows: (i) we adapt a previously developed IPP approach based on IG estimation [13], replacing the theoretical sensor model with an empirical model based on sampling real-world image data; (ii) we determine in simulation how the IPP approach and the resulting mapping error are influenced by the level of sampling effort; (iii) we validate our approach using imagery from a real-world wheat field, demonstrating its effectiveness in improving the accuracy of the lodging map. In Sect. 2, we present the IG-based IPP approach to field mapping, which represents the basis of our study. In Sect. 3, we introduce the empirical sampling-based model. In Sect. 4, we discuss the results for both simulated and real wheat fields. Section 5 concludes by addressing the necessary future steps for deployment in the field. Code and details about the experimental setup are available at https://github.com/AGR-o-RAMA/active_sensing.

2 Problem Statement and IPP Approach

Following previous work [13], we use an occupancy map to represent the wheat field, where each cell indicates whether the corresponding area is lodged or not.

A single UAV equipped with a downward-facing camera navigates this environment, collecting observations to update the occupancy map to compute the probability of lodging for each cell, and making informed decisions about the next actions. The goal is to minimise the mapping error and flight time/energy consumption, enabling efficient and precise mapping of lodged areas.

2.1 Environment

The agricultural terrain is represented as an occupancy grid, that is, a set $\mathcal{M}$ of equally sized cells (e.g., squares of 1 m). Each cell $i \in \mathcal{M}$ corresponds to a specific area of the field, is characterised by a state $m_i \in \{0, 1\}$ (respectively, not lodged or lodged) and is associated with an estimated probability of lodging $P(m_i = 1)$. The UAV operates in a 3D lattice of waypoints above the terrain, as described in previous work [13]. The UAV has accurate knowledge of its position (for instance, exploiting RTK-GNSS technology) and can choose among

a limited set of actions to move towards a close waypoint. As the UAV navigates, its downward-facing camera captures observations within its footprint (assumed without loss of generality to be square and characterized by the field of view F). The footprint at position $\mathbf{x}_t = (x_t, y_t, h_t)$ is denoted as $C(\mathbf{x}_t) \subset \mathcal{M}$ and has size $d(h_t) = 2h_t \tan(F/2)$, where h_t is the altitude of position $\mathbf{x}_t$. Each observation $\mathbf{o}_t = (\mathbf{x}_t, \mathbf{z}_t)$ consists of the position of the UAV $\mathbf{x}_t$ and sensor measurements for each observed cell $\mathbf{z}_t = \{z_{i,t} | i \in C(\mathbf{x}_t)\}$. Each observation is used to update a local belief map about the probability of lodging for each cell. We assume the availability of a sensor model $P(z_{i,t} \mid m_i, \mathbf{x}_t)$ which provides the probability of having the observation $z_{i,t}$ from position $\mathbf{x}_t$, given the true condition of cell m_i.

2.2 Mapping

To capture spatial correlation among cells, the belief map is modelled as a Conditional Random Field (CRF) [16], structured as a rectangular lattice graph $G = (\mathcal{M}, \mathcal{E})$, where each node $i \in \mathcal{M}$ corresponds to a terrain cell, and each edge $(i, j) \in \mathcal{E}$ models the spatial dependencies between adjacent cells.

At time t, when observation $\mathbf{z}_t$ is made, the probabilistic belief map is updated through a three-step process:

1. **Bayesian update**: The belief of each cell in the footprint is updated using Bayes' theorem:

$$\forall m \in \{0,1\} \quad P(m_i = m | z_{i,t}) = \frac{P(z_{i,t} \mid m_i = m, \mathbf{x}_t) P^-(m_i = m)}{P(z_{i,t})}, \tag{1}$$

$$P(z_{i,t}) = \sum_{m \in \{0,1\}} P(z_{i,t} \mid m_i = m, \mathbf{x}_t) P^-(m_i = m) \tag{2}$$

   where the probability $P(m_i = m | z_{i,t})$ is obtained by updating the prior probability $P^-(m_i = m)$ on the basis of the observation $z_{i,t}$, the sensor model $P(z_{i,t} \mid m_i = m, \mathbf{x}_t)$ and the marginal probability $P(z_{i,t})$ of observing $z_{i,t}$.
2. **Belief Propagation**: Local updates are propagated to neighbouring cells to maintain consistency throughout the entire belief map by using the Sum-Product variant of the Loopy Belief Propagation (LBP) algorithm [11]. This is an iterative message passing process: a cell i sends two messages $\mu_{i \to j}(\hat{m}_j)$ to its neighbour $j \in \mathcal{N}(i)$, one for each possible state $\hat{m}_j \in \{0, 1\}$:

$$\mu_{i \to j}^{(q+1)}(\hat{m}_j) = \sum_{m \in \{0,1\}} P(z_{i,t} \mid m_i = m, \mathbf{x}_t) W_{ij}(\mathbf{o}_t) \prod_{\substack{k \in \mathcal{N}(i) \\ k \neq j}} \mu_{k \to i}^{(q)}(m), \tag{3}$$

   where $W_{ij}(\mathbf{o}_t)$ represents the pairwise factor of the CRF, which depends on m_i, $\hat{m}_j$ and $\mathbf{o}_t$ (see below). The process starts with $\mu_{i \to j}^{(0)}(\hat{m}_j) = 0.5$ for all $i, j \in \mathcal{M}$, and continues until convergence or for a fixed number of iterations q_{max}. In this work, we limit to a single loop, hence $q_{max} = 1$.

3. **Marginalization**: Final occupancy probabilities—the posterior $P^+(m_i)$—are computed by marginalising all messages:

$$\forall m \in \{0,1\} \quad P^+(m_i = m) = \frac{1}{Z} P(m_i = m | z_{i,t}) \prod_{j \in \mathcal{N}(i)} \mu_{j \to i}(m), \tag{4}$$

where Z is a normalisation factor ensuring that $P(m_i) \in [0,1]$.

The sensor model $P(z_{i,t} \mid m_i, \mathbf{x}_t)$ maps the observed measurements to the state of individual cells, which are the nodes $\mathcal{M}$ of the CRF. It plays a critical role in the belief update process, assigning a likelihood to observations based on the sensor characteristics and the observation position, and accounts for potential errors—e.g., false positives (FP) and false negatives (FN). This allows the system to weigh observations appropriately, ensuring that less reliable data (e.g., observations at very high altitudes) have minimal influence on the belief map.

The pairwise factors of the CRF are associated with the edges $\mathcal{E}$, modelling spatial dependencies between neighbouring cells. $W_{ij}(\mathbf{o}_t)$ quantify the strength of spatial correlations, reflecting how the state of one cell influences its neighbours. We adopt three pairwise factor weighting approaches, as described in [13]:

- **Equal**: Assigns a uniform weight of $W_{ij}(\mathbf{o}_t) = 0.5$ to all pairs, disregarding spatial correlations.
- **Biased**: Encourages local consistency by assigning a higher weight when neighbouring cells are thought to share the same state: $W_{ij}(\mathbf{o}_t) = 0.7$ if $m_i = m_j$, and 0.3 otherwise.
- **Adaptive**: Dynamically adjusts weights based on the estimated spatial correlation within the current camera footprint $C(\mathbf{x}_t)$. Estimation is performed on each cell of the footprint and its Von Neumann neighbourhood, where c is the value of the central cell and n is the sum of values from its four neighbouring cells. The correlation is computed using the Pearson correlation coefficient $\hat{\rho}_{c,n} \in [-1,1]$ [4], followed by a normalisation step via a sigmoid function. The resulting weight $W_{ij}(\mathbf{o}_t)$ is in the range $[0.27, 0.73]$, depending on the estimated correlation coefficient. For details, see [13].

2.3 Planning

Once new observations are mapped into the belief map, the path planning algorithm determines the next best action to take. Optimal actions prioritise observations of areas with high uncertainty, ensuring efficient and accurate mapping. More specifically, the UAV selects actions to maximise the information gain (IG), defined as the expected reduction in uncertainty of the belief map.

At each step, the UAV evaluates the valid actions $a \in A(x_t) \subseteq \mathbf{A} = \{\text{up}, \text{down}, \text{front}, \text{back}, \text{left}, \text{right}, \text{hover}\}$, considering that some actions may not be available when the UAV is at the boundaries of the 3D lattice.

For each candidate action a, the UAV considers the destination $\mathbf{x}_{t+1}(a)$ and simulates a belief update $P(m_i|\hat{z}_i)$ for cells within the footprint $C(\mathbf{x}_{t+1}(a))$ with each new possible observation $\hat{z}_i = z_{i,t+1}(a) \in \{0,1\}$, following Eq. (1).

Then, the IG for a cell i is calculated as the difference between its current entropy $H(m_i)$ and its expected entropy $H(m_i \mid \mathbf{x}_{t+1}(a))$:

$$\mathrm{IG}(m_i, \mathbf{x}_{t+1}(a)) = H(m_i) - H(m_i \mid \mathbf{x}_{t+1}(a)), \tag{5}$$

where

$$H(m_i) = -\sum_{m \in \{0,1\}} P^+(m_i = m) \log(P^+(m_i = m)), \tag{6}$$

$$H(m_i \mid \mathbf{x}_{t+1}(a)) = \sum_{z \in \{0,1\}} P(\hat{z}_i = z) H(m_i | \hat{z}_i = z), \tag{7}$$

$$H(m_i|\hat{z}_i) = -\sum_{m \in \{0,1\}} P(m_i = m|\hat{z}_i) \log(P(m_i = m|\hat{z}_i)), \tag{8}$$

where $P(\hat{z}_i = z)$ is computed with Eq. (2). The total IG for an action a is computed by summing the IG values for all cells in $C(\mathbf{x}_{t+1}(a))$, and represents the overall reduction in uncertainty expected from executing action a:

$$\mathrm{IG}(a) = \sum_{i \in C(\mathbf{x}_{t+1}(a))} \mathrm{IG}(m_i, \mathbf{x}_{t+1}(a)). \tag{9}$$

The UAV then selects the optimal action a^* by choosing the action that maximizes the total IG:

$$a^* = \arg \max_{a \in A(\mathbf{x}_t)} \mathrm{IG}(a). \tag{10}$$

In cases where multiple actions yield the same highest IG value, a random one is chosen to break the tie. Altitude h_t influences the trade-off between footprint size and observation accuracy: observations from higher altitudes cover more cells but the error rates increase, while lower altitudes provide more precise observations over smaller areas. The planning algorithm maximises the overall IG by dynamically adjusting the altitude to balance these factors.

3 Sensor Model

As mentioned in the previous sections, the sensor model is a critical component of the UAV's IPP framework, as it determines how observations update the belief map and how the IG is computed. While a theoretical representation based purely on altitude-dependent assumptions is useful to assess the performance of the IPP approach, a model must be defined to represent real-world observations. In the following section, we present (i) the theoretical altitude-based sensor model; (ii) a representation exploiting altitude-dependent classification quality representation; and (iii) the derived empirically sampled sensor model.

3.1 Theoretical Sensor Model

The theoretical sensor model characterizes measurement accuracy by defining the probability of observing a specific measurement $z_{i,t}$ given the true state m_i of a cell and the UAV's position $\mathbf{x}_t$:

$$p(z_{i,t} \mid m_i, \mathbf{x}_t) = \begin{cases} 1 - \sigma(\mathbf{x}_t) & \text{if } z_{i,t} = m_i, \\ \sigma(\mathbf{x}_t) & \text{if } z_{i,t} \neq m_i. \end{cases} \tag{11}$$

where $\sigma(\mathbf{x}_t)$ denotes the probability of incorrect measurements at position $\mathbf{x}_t$. Since the dominant factor affecting UAV-based image quality is altitude, this error probability is commonly simplified to a function depending solely on h_t:

$$\sigma(\mathbf{x}_t) = \min(0.5, a(1 - e^{-bh_t})), \quad \mathbf{x}_t = (x_t, y_t, h_t) \tag{12}$$

where a and b are parameters which can be obtained from literature or from preliminary experimentation. Note that $\sigma(\mathbf{x}_t)$ is capped at 0.5, as this value corresponds to a purely random choice, i.e., the worst possible sensor.

3.2 Sensor Model as a Confusion Matrix

For field deployment, the actual classification of lodged and non-lodged areas from UAV imagery should be taken into account. Here, increasing the UAV altitude leads to a higher ground sampling distance (GSD), which reduces image resolution and classification accuracy. Conversely, lower altitudes result in finer GSD, yielding higher image resolution and more reliable observations. A theoretical sensor model with a single parameter set (a, b) for both lodged and non-lodged states assumes symmetric error rates, an assumption that may not hold true in practice. Also, varying weather or field conditions can significantly affect false-positive and false-negative error rates, necessitating distinct altitude-specific representations of sensor accuracy. To explicitly capture realistic sensor performance at different altitudes, the sensor model can be represented as an altitude-dependent confusion matrix (see Table 1). In this formulation, the false positive rate $\sigma_0(\mathbf{x}_t)$ and the false negative rate $\sigma_1(\mathbf{x}_t)$ are altitude-dependent values. For each altitude level, a unique confusion matrix can be constructed, reflecting the sensor's performance at that specific height, and can be directly integrated in the mapping and planning phase as follows:

$$p(z_{i,t} \mid m_i, \mathbf{x}_t) = \begin{cases} 1 - \sigma_0(\mathbf{x}_t) & \text{if } z_{i,t} = m_i = 0, \\ \sigma_0(\mathbf{x}_t) & \text{if } z_{i,t} = 1,\ m_i = 0, \\ 1 - \sigma_1(\mathbf{x}_t) & \text{if } z_{i,t} = m_i = 1, \\ \sigma_1(\mathbf{x}_t) & \text{if } z_{i,t} = 0,\ m_i = 1, \end{cases} \tag{13}$$

Representing sensor performance explicitly as a confusion matrix enables the direct integration of realistic, altitude-specific classification uncertainties into the UAV's information gain-based path planning framework, facilitating more effective and practical decision-making.

Table 1. Sensor model represented as a confusion matrix; the parameters $\sigma_0(\mathbf{x}_t)$ and $\sigma_1(\mathbf{x}_t)$ represent different false-positive and false-negative error rates in the classification, varying with the altitude h_t associated with position $\mathbf{x}_t$

	$m = 0$	$m = 1$
$z = 0$	$1 - \sigma_0(\mathbf{x}_t)$	$\sigma_1(\mathbf{x}_t)$
$z = 1$	$\sigma_0(\mathbf{x}_t)$	$1 - \sigma_1(\mathbf{x}_t)$

3.3 Sampled Sensor Model

While theoretical confusion matrices provide a structured representation of sensor performance, their parameters are often idealized and may not fully reflect real-world sensor behaviour. Thus, we propose a sampling-based approach to empirically derive these matrices from actual drone observations. This method replaces theoretical assumptions with experimentally obtained data, capturing practical sensor characteristics.

To construct the empirical confusion matrix, we collect UAV observations at different altitudes and calculate altitude-specific false positive (σ_0) and false negative (σ_1) rates. The required number of samples N for reliable estimates can be determined considering a Bernoulli process. In this context, each sample is a Bernoulli trial with two possible outcomes: correct or incorrect observation (depending on the cell type, either "lodged" or "not lodged"). To estimate the required number of samples given a 95% confidence level and error margin E, we refer to the sample size of a binomial proportion estimator:

$$N = \frac{Z_s^2 \, p(1-p)}{E^2} \tag{14}$$

where N is the required number of independent trials (i.e., labelled samples), p is the probability of success of a single Bernoulli trial, E is the tolerated margin of error, and $Z_s = 1.96$ is the Z-score corresponding to a 95% confidence level reflecting the desired level of statistical certainty. As p is not known beforehand for a real field, we make the conservative choice of setting $p = 0.5$. This choice maximizes the product $p(1-p)$ and thus yields the largest required sample size. This ensures that the empirical estimates of the confusion matrix entries (i.e., error rates) are statistically robust, thereby supporting the reliability of the resulting sensor model used for informative path planning.

4 Experimental Results

To evaluate the proposed IG-based IPP approach with the empirical sensor model, we first assess the mapping accuracy and efficiency in simulation, varying the sampling effort according to desired error margins. This analysis aims at revealing how robust the IPP approach is with respect to imprecise sensor models. Following this, we showcase the IPP approach exploiting real-world imagery of a wheat field.

4.1 Simulated Field

Simulated wheat fields are generated using a Gaussian random field that covers an area of 50 m × 50 m, discretised into 400 × 400 cells with a 12.5 cm side. The spatial correlation length is set to $r = 4$, producing random fields with well-distinguishable patches in which lodging occurs (see Fig. 1 left). The UAV configuration space is also discretised, with horizontal waypoints spaced approximately 3 m apart, and vertical waypoints spanning altitudes from 5.4 m to 32.4 m in increments of 5.4 m (six levels in total). The camera field of view F is set to $\frac{\pi}{3}$. We ensure that the observation footprint intersects the monitored area near boundaries. Any out-of-bounds image portion is discarded. The UAV starts at a random position along the field boundaries.

We consider a theoretical sensor model with $a = 1$ and $b = 0.015$, which constitutes our baseline. To implement the sampled sensor model, we determine the necessary sample size N for each altitude according to three different error margins: (i) low error and high cost $E = 0.05$; (ii) medium error and moderate cost $E = 0.1$; (iii) high error and low cost $E = 0.3$. From the sensor model in (11), the desired sample size can be calculated for each altitude of the UAV configuration space, as shown in Fig. 1 right.

We consider the three different approaches for pairwise factor weighting and the four sensor models (baseline plus three error margins). For each combination, we run 20 independent runs in which the UAV tries to map a randomly generated field. At the start of each run, a suitable number of samples N is collected for each altitude h, determined by the error margin E. These samples are then used to construct the empirical confusion matrix, replacing the theoretical sensor model. In other words, each run is characterized by slightly different confusion matrices resulting from the independent sampling process. In this way, we can also appreciate how much variability is induced by the empirical sensor model in the mapping process. To evaluate the mapping outcome, we measure the residual entropy of the belief map and the mean squared error (MSE) with respect to the ground truth. Additionally, to better appreciate the IPP outcomes, we also record the height of the UAV while mapping and the percentage of field coverage.

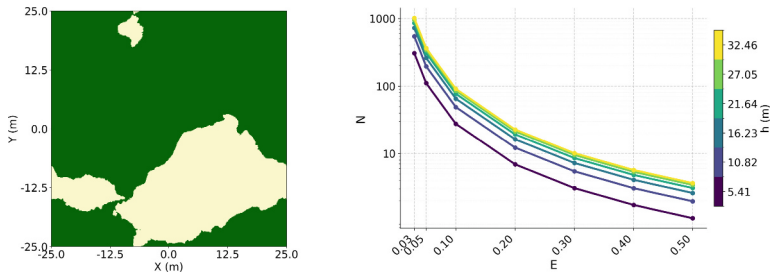


Fig. 1. Left: Gaussian random field generated with $r = 4$, simulating a wheat field with lodging; right: required sample sizes computed from the theoretical sensor model ($a = 1$, $b = 0.015$) across different altitudes and precision levels

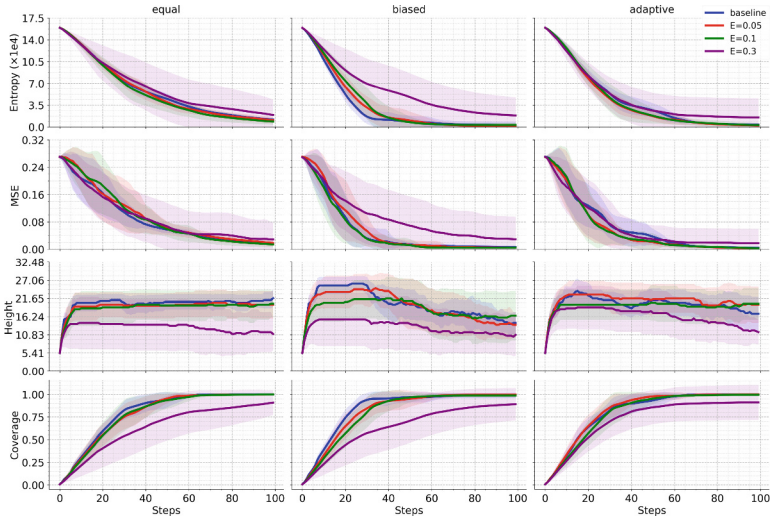


Fig. 2. Performance metrics for the theoretical sensor model (baseline) and sampled models ($E = 0.05, 0.1, 0.3$) on a Gaussian field with equal, biased, adaptive weights (left to right); shaded areas indicate standard deviation and solid lines the mean over 20 independent runs

The experimental results obtained from simulations are shown in Fig. 2. For the theoretical model (baseline), the mean entropy and MSE exhibit stable convergence over time, with minimal standard deviation, reflecting consistent performance. In contrast, the usage of sampled sensor models introduces higher variability, as evidenced by larger standard deviations due to the variable confusion matrix: the higher the error margin, the higher the standard deviation. Despite this variability, the mean entropy and MSE for the sampled models converge to values comparable to those of the theoretical model, with significant deviations visible only for the largest error margins $E = 0.3$. This indicates that the sampled approach effectively balances adaptability to real-world data and accuracy. Notably, for adaptive weights, standard deviations are smaller and mean values are closer to the theoretical values even for fewer samples that correspond to $E = 0.3$. This indicates an ability of the adaptive approach to compensate for errors introduced by sampling by suitably tuning the CRF weights.

The mapping behaviour and performance are consistent with previous work [13]. Specifically, the high spatial correlation of the Gaussian random field ($r = 4$) entails better performance of the biased CRF weights over the equal CRF weights, with a faster reduction of entropy and better mapping accuracy. The adaptive approach leads to similar performance, as it can easily tune the CRF weights to the field conditions. The height and coverage dynamics further highlight the trade-offs introduced by sampling. While the sampled models exhibit greater fluctuations in altitude and coverage due to higher standard deviation, their mean values align closely with the theoretical model. Height varies

significantly with the weighting strategy. For equal weights, the theoretical and sampled models ($E = 0.05$, $E = 0.1$) maintain relatively stable and similar altitudes, while the high error introduced when the sampling effort is low ($E = 0.3$) requires monitoring at lower altitudes, on average.

The results clearly demonstrate that sampled sensor models effectively balance accuracy and flexibility. Specifically, smaller error margins achieve performance comparable to the theoretical model while accommodating realistic sensor variability. Larger error margins lead to increased variability, which can, however, be partially compensated for by the adaptive CRF weights.

4.2 Wheat Field

To validate the proposed sensor models under real-world conditions, we conducted experiments using UAV imagery captured over an actual wheat field. These images have been acquired in May 2024 close to Adria in Italy using a DJI Mavic 3 Pro equipped with a 10MP RGB camera, flying at 19.5 m altitude. From such imagery, we built an orthomap and divided it into discrete cells of size 1 m × 1 m. The cells belonging to the wheat field were then extracted (6600 cells, each of 189 × 189 pixels) and manually labelled as either "lodged" or "not lodged" (see Fig. 3).

The binary classification of wheat lodging was performed for each cell using a CNN adapted from *MobileNetV3-Small* pretrained on ImageNet1000 [6]. Specifically, we modified its final layers to suit our binary classification task, employing a linear layer with 128 output features, followed by ReLU activation, a dropout layer with $P = 0.4$, and a final linear layer for binary classification. Out of the 6600 cells, 70% were used for training, 15% for validation, and 15% for testing. After training, the resulting model reached 96% in accuracy and F1 score, with only limited false-positive and false-negative rates (respectively, 7.43% and 2.66%).

To simulate a UAV flying over the wheat field, we define the UAV configuration space to maintain approximately 80% forward and side overlap between adjacent images, resulting in a horizontal way-point spacing of 4.5 m. Consistent with the Gaussian random field scenario, we derived the flight altitudes from

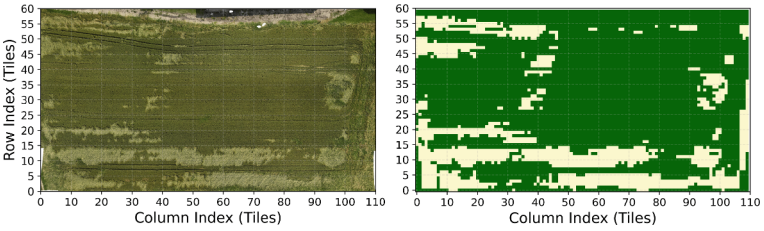


Fig. 3. Left: orthomap of the wheat field divided into 1 m × 1 m cells; right: manual annotation showing lodged (light yellow) and not lodged (dark green) areas

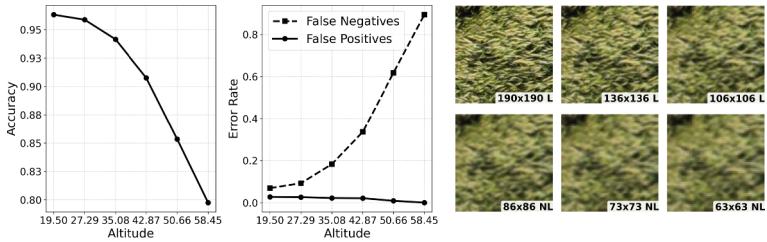


Fig. 4. Left and center: classification accuracy and FP, FN rates as functions of altitude, showing the trade-off between footprint size and image resolution; Gaussian blur simulates the image quality degradation at higher altitudes; right: example tile (47, 55) at varying altitudes, illustrating the effect of resolution on classification, with crop size in pixels and predicted class label (L—lodged, NL—not lodged) shown for each altitude

the field of view F, resulting in altitude increments of 7.79 m. This configuration yielded six discrete altitude levels ranging from 19.5 m to 58.45 m.

To simulate the effect of altitude changes on classification performance, a Gaussian blur was applied to simulate reduced image resolution at higher altitudes, representing realistic sensor degradation due to increasing GSD. As illustrated in Fig. 4, classification accuracy declines from approximately 96% at 19.5 m to around 80% at 58 m. This degradation is further characterized by changes in the false positive (FP) and false negative (FN) rates: the FN rate increases from about 0.07 at 19.5 m to over 0.89 at 58.45 m, while the FP rate decreases from roughly 0.03 to 0.004 over the same range.

This divergence arises from a combination of image degradation and the distribution of wheat conditions in the field, as shown in Fig. 3. At higher altitudes, blurring and pixel downsampling cause lodged regions—often characterized by thin, yellowish patches or irregular textures—to blend into surrounding healthy vegetation. These fine-scale patterns become indistinguishable, leading the classifier to incorrectly label lodged tiles as "not lodged," thus increasing FN rates. Conversely, non-lodged areas remain visually consistent and are less affected by altitude, causing false positives to decrease. This asymmetry indicates that classification uncertainty becomes skewed toward missed detections (FN) rather than false alarms (FP) as resolution drops.

Given the above classification model, we obtain the empirical model by sampling cells from the real field. Specifically, we calculate the required number of samples N as mentioned above, randomly select N cells from the orthomap, and simulate predictions at different altitude levels. Using these predictions and the corresponding annotations, we construct confusion matrices for each altitude. Also in this case, we consider the three approaches for the CRF weights and three different levels of error margin ($E = 0.05, 0.1, 0.3$). For each configuration, we perform 20 independent runs, each time recomputing a new sampled sensor model and launching the UAV from a different random position along the field boundary. Given that we cannot rely here on a theoretical sensor model, as a

baseline we consider a standard lawnmower strategy wherein the UAV sequentially covers the whole field flying at the lowest altitude (19.5 m). We consider for the baseline a perfect classifier, directly accessing the ground truth value for each cell under the UAV footprint. The results of our simulations are shown in Fig. 5. Despite the assumed perfect knowledge of the field, the baseline only achieves about 35% coverage in the limited allotted time. Hence, the residual entropy and MSE remain high due to the effect of non-visited areas. The usage of sampled sensor models enables exploiting the IPP approach, consistently outperforming the baseline, achieving quicker entropy and MSE reductions and higher coverage rates. The MSE consistently decreases but never reaches zero due to inherent classification errors introduced by the trained classifier. In this validation, we observe a lower impact of the error margin on the performance, which on average is quite similar. The standard deviation remains high for the highest error margin ($E = 0.3$), revealing that, also with real-world data, low sampling efforts increase the variability of the mapping performance. Different weighting strategies influenced the sampled models' performance. Adaptive weighting strategies provided the best stability and most efficient altitude adjustments, while biased weighting showed notable altitude fluctuations due to increased uncertainty. Equal weighting represented a balanced approach but offered less adaptability compared to the adaptive strategy, especially for the case of low sampling effort. Overall, these experiments validate the significant advantages of employing empirically sampled sensor models in UAV-based precision agriculture. Specifically, the adaptive CRF weighting approach offers a highly effective combination, optimizing accuracy, robustness, and operational efficiency compared to traditional, rigid mapping approaches.

5 Conclusions

In this paper, we have presented an approach for information gain-based UAV informative path planning applied to precision agriculture, specifically targeting wheat lodging detection. Our key contribution lies in transitioning from traditional altitude-based theoretical sensor models to empirically derived sensor models using practical UAV observations. Through evaluation on both simulated Gaussian random fields and a real-world field, we demonstrated significant advantages of empirically derived sensor models, laying a path toward actual deployment in the field.

The proposed approach requires minimal sampling of the field before starting the mapping. We posit that this can be done by collecting just a few images, possibly only a single image for each altitude to be considered. This allows obtaining a sufficient number of cells under the camera footprint, limiting the manual annotation effort. This process may also exploit additional images collected and annotated to fine-tune the classification model to the specific weather and field conditions, which is often a requirement for field operation. Hence, the sampled sensor models can be obtained at practically no additional cost.

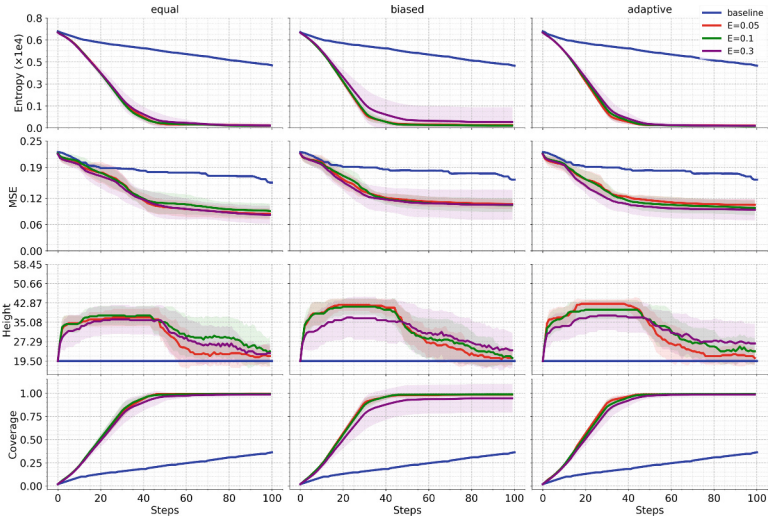


Fig. 5. Performance metrics for the lawnmower strategy (baseline) and sampled models ($E = 0.05, 0.1, 0.3$) on an orthomap field with pairwise equal, biased, and adaptive weights; shaded areas indicate standard deviation, and solid lines indicate mean values

The accuracy of the classification model is an important aspect, but note that this model must be small enough to be deployed onboard the UAV. For the IPP approach to be executed, the classification model should be good enough to support planning and collect more observations where the uncertainty is higher. To improve mapping, the imagery resulting from this activity can be analysed offline and off-board by larger and more accurate classification models, hence providing better information for operational and strategic decision making.

Acknowledgments. This study was supported by the AGRITECH project (PNRR, Italian Ministry of University and Research; CUP CN00000022).

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